

Updated wheat seed rate and spacing

The wheat seed-rate and spacing figures in this paper have been superseded by the figures locked in the PQNK book, Chapter 29 (SIPP and Precision Planting).

Wheat is sown in 5 rows on the 42-inch bed at 8-inch seed-to-seed spacing: about 62,000 hills per acre, with 2 seeds per hill (about 124,000 seeds per acre). The recommended seed rate is 8 kg per acre for the first crop, reduced season by season toward the calculated rate of under 5 kg per acre on a mature, precisely planted farm. The 2.64 kg per acre recorded in the NARC trial sits at the most experienced end of that same progression. Seed size runs from about 10,000 seeds per kg in long, cool grain-filling regions to about 28,000 seeds per kg in Punjab, so always calibrate with the actual seed lot.

Where this paper gives a precision seed rate of 10 kg per acre (including in its seed-rate and cost tables), use the figures above; the lower first-crop and mature rates only reduce the seed cost shown.

The original paper follows unchanged.



Knowledge Paper: Transforming Wheat Production through the Paedar Qudratti Nizam Kashatqari (PQNK)

Title: The PQNK Pathway to Doubling Farmers' Income and Dominating the Organic Wheat Export Market



1. Preamble: The PQNK Lens

The Paedar Qudratti Nizam Kashatqari (PQNK) is not merely a set of agricultural practices; it is a paradigm shift. It is a philosophy that views the farm as a self-regenerating, pristine ecosystem. The PQNK lens moves us away from the extractive and input-intensive model of agriculture, which has depleted soils, depleted water tables, and depleted farmer prosperity. Instead, PQNK works in harmony with nature's intelligence, fostering soil biology to unlock unparalleled productivity, sustainability, and profitability. This paper applies the PQNK lens to wheat production, revealing a pathway to national prosperity.

2. The Problem: The Unsustainable Conventional Wheat Cycle

Conventional wheat farming is trapped in a cycle of high costs and diminishing returns.

- **High Input Costs:** Farmers are burdened by the costs of repeated tillage, synthetic fertilizers (DAP, Urea), weedicides, and frequent irrigation.

- **Stagnant Yields:** The average yield often plateaus around 30 maunds per acre, unable to compensate for rising input costs.
- **Soil Degradation:** Chemical fertilizers and tillage destroy soil organic matter and beneficial microbial life, leading to long-term fertility loss.
- **Low Profit Margins:** The combination of high costs and a commodity-market price keeps farmer incomes low and vulnerable.

3. The PQNK Solution: A Symphony of Regenerative Practices

PQNK for wheat is a precise, knowledge-based system that dramatically reduces external inputs while boosting yield and quality.

4. The Economic Transformation: A Comparative Analysis (Per Acre)

Let us translate these practices into a clear financial picture for one acre of wheat.

Assumptions:

- **Conventional Yield:** 30 Maunds/Acre
- **PQNK Yield:** 45 Maunds/Acre
- **Conventional Wheat Price:** PKR 3,500 /maund (Approx. Govt. Support Price)
- **PQNK Wheat Price:** PKR 7,500 /maund (Conservative Intl. Organic Price)

Cost of Production (Per Acre Estimate)			
Cost Component	Conventional System (PKR)	PQNK System (PKR)	Saving with PQNK (PKR)
Tillage & Sowing	8,000	2,500 (Direct Seeding)	5,500
Seed	6,000 (60kg)	1,000 (10kg)	5,000
Fertilizers (DAP, Urea)	18,000	0	18,000
Weedicides	3,000	0	3,000
Irrigation	4,000	1,500	2,500
Total Operational Cost	39,000	5,000	34,000 (87% Less)

Revenue & Profit (Per Acre)		
Metric	Conventional System (PKR)	PQNK System (PKR)
Yield	30 Maunds	45 Maunds
Price per Maund	3,500	7,500
Gross Revenue	105,000	337,500
Cost of Production	39,000	5,000
Net Profit	66,000	332,500

Comparison between Conventional and PQNK			
Practice	Conventional System	PQNK System	PQNK Impact
Land Preparation	3-4 tillage operations	Zero Tillage. Direct seeding into soil with previous crop residue.	Saves fuel, labour, time, and preserves soil structure and moisture.
Seed Rate	60 kg/acre (dense sowing)	10 kg/acre (precision sowing)	Saves 50 kg/acre seed cost. Plants are healthier, less prone to disease.
Fertilization	1x 50kg DAP Bag + 2x Urea Bags	None. Soil biology provides nitrogen and solubilizes phosphorus from soil reserves and all other nutrients	Eliminates the single largest cost component. Builds natural soil fertility.
Weed Management	Reliance on chemical weedicides	No Weedicides. Managed by residue mulch, and soil health.	Eliminates cost, removes chemical burden from the ecosystem.
Irrigation	4-5 flood irrigations	Maximum 2 Furrow Irrigations. Residue mulch drastically reduces evaporation.	Saves 70-80% water, crucial for water-scarce regions. Reduces energy cost.
Yield	~30 Maunds/Acre	~45 Maunds/Acre	Healthier plants in a vibrant ecosystem express their full genetic potential.
Product Quality	Standard Commodity Wheat	Pristine Organic Plus Wheat	Chemical-free, nutrient-dense, superior baking quality.

4A. The Foreign Exchange Opportunity: A 20% Conversion Scenario

To understand the immediate opportunity, let us project the impact of a conservative and achievable target: converting 20% of the nation's total wheat area to the PQNK system.

Scenario Assumptions:

Total National Wheat Area: ~ 9.0 million hectares (approx. 22.2 million acres)

Area under PQNK (20%): 1.8 million hectares (approx. 4.44 million acres)

PQNK Yield: 4.0 Tonnes/Hectare (~ 45 Maunds/Acre)

Total PQNK Production: 1.8 million hectares 4.0 T/Ha = 7.2 million tonnes

Exportable Surplus: Assuming 80% of PQNK production is exported for premium forex earnings = 5.76 million tonnes

Export Price: A conservative \$650/tonne for "Organic Plus" wheat.

Foreign Exchange (Forex) Earnings Calculation:

Total Forex Earnings = Exportable Surplus × Export Price

Total Forex Earnings = 5.76 million tonnes × \$650/tonne

Total Forex Earnings = US \$3.744 Billion

This figure of nearly \$3.75 Billion is not merely an increase in revenue; it is a strategic infusion of high-value foreign exchange earned from a sector traditionally struggling with low margins. It represents a paradigm where agriculture becomes a leading export sector, not a subsidized burden.

Comparative Impact:

This \$3.75 billion from just 20% of the wheat area under PQNK would be earned by selling a premium, value-added product. It could potentially rival or surpass forex earnings from certain commodity sectors that have far higher production and export complexities.

Conclusion of the Scenario:

A 20% adoption of PQNK presents a low-risk, high-reward pathway to generating substantial forex reserves. It builds a powerful case for prioritizing agri-exports as a key pillar of national economic strategy, starting with this readily achievable target.

Conclusion: The PQNK system increases net profit per acre from PKR 66,000 to PKR 332,500. This is an increase of over 400%. The farmer's income is not just improved; it is revolutionised.

5. National Scale Impact: The 50% PQNK Adoption Scenario

If 50% of the national wheat area is brought under PQNK production:

1. **Farmer Prosperity:** Millions of farming families will see their per-acre profit increase by PKR 266,500. This transitions them from subsistence to genuine prosperity.
2. **Export Boom:** Pakistan becomes a global powerhouse for high-value "Organic Plus" wheat, commanding a premium price in the international market and earning crucial foreign exchange.
3. **Environmental Resilience:**
 - Water Saved: Billions of cubic meters of irrigation water are conserved.
 - Soil Revived: Millions of acres of farmland begin to regenerate.
 - Chemical Halt: The environment is cleansed of thousands of tonnes of synthetic fertilizers and pesticides.
4. **Food Security:** With a 50% increase in yield per acre on half the area, total national wheat production would see a significant net increase, enhancing food security.

6. The Proposal: A Call to Action

We propose a national mission, "PQNK Wheat for Prosperity," with the following pillars:

- **Pillar I: Massive Knowledge Transfer:** Train a cadre of PQNK master trainers and establish showcase training farms in every district.

- **Pillar II:** Market Linkage & Branding: Create a "PQNK Organic Plus" brand, directly link farmer producer organizations (FPOs) with international buyers, and develop robust supply chains for exports.
- **Pillar III:** Policy Support: Initially, provide support for the transition, such as access to set of PQNK machines. The system's own economics will make it self-sustaining thereafter.
- **Pillar IV:** Research & Development: Establish a PQNK production management entity to manage programmed production to achieve export targets.

7. Conclusion

The PQNK system for wheat is a ready-now solution. It is a win for the farmer, a win for the national economy, and a win for the planet. By choosing PQNK, we are not just growing a crop; we are cultivating a future of abundance, resilience, and pristine health. Let us embrace this wisdom and embark on this transformative 'picnic' for our nation.

Glossary:

- **PQNK:** Paedar Qudratti Nizam Kashatqari - The Regenerative & Sustainable Pristine Organic Farming System.
- **Organic Plus:** Denotes produce that not only is certified organic but is grown in a way that actively enhances the ecosystem, going beyond standard organic definitions.
- **Maund:** A traditional unit of mass used in South Asia, equal to 40 kilograms.

Footnote: *Any Production Process That Inundates Soil With Water, Disturbs Soil Through Tillage, Or Leaves Soil Bare Without Organic Mulch Cover Does Not Qualify As Natural Ecosystem Science For Production Agriculture.*

PQNK, to be pronounced as 'picnic', which stands for Paedar Qudratti Nizam Kashatqari, and means: the regenerative & sustainable Pristine Organic Farming System.

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<https://www.youtube.com/@pedaverpqnk3167/videos>
pedaver@gmail.com
 WhatsApp +92 320 677 6666